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(54) **BLUEBERRY PLANT VARIETY NAMED**
‘BLUECSOL7’

(50) Latin Name: *Vaccinium corymbosum* L.
Varietal Denomination: **BLUECSOL7**

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patent is extended or adjusted under 35
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(58) **Field of Classification Search**
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(57) **ABSTRACT**

A new and distinct evergreen blueberry variety with zero
chill requirement named ‘BLUECSOL7’, characterized by
the following combination of traits. Strong plant vigor,
semi-upright growth habit, ovate leaf shape, globose corolla
shape and weak conspicuousness of ridges on corolla tube,
oblate fruit shaped, extra-large fruit-size, moderate bloom
intensity, medium sweetness, high acidity, and very firm
fruit. Latin name of the genus and species: *Vaccinium*
corymbosum L.

6 Drawing Sheets

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Latin name: *Vaccinium corymbosum* L.
Variety denomination: Blueberry plant named
‘BLUECSOL7’.

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to a Peruvian Breeder’s
Rights Application No. 55-2022 filed on Jan. 12, 2022, the
entire contents of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

The present invention relates to a new *Vaccinium corym-*
bosum L. plant, which was selected from open pollination
between ‘Biloxi’ (*Vaccinium corymbosum* L., public plant)
as the female (seed) parent and pollen from an unknown
variety (*Vaccinium corymbosum* L.). Berry seeds of plants of
the ‘Biloxi’ were collected from different fields located in
Viru, La Libertad, Peru at the beginnings of 2016. 11805
seedlings at high density were planted under field conditions
in December of the same year.

The seedlings were evaluated during 3 growing seasons
(2017, 2018 and 2019) and the breeding method used was
phenotypic recurrent selection. In December 2019 one of the
seedlings, ‘BLUECSOL7’, was isolated due to its excellent
quality fruit, long postharvest life, and high yield potential.
‘BLUECSOL7’ was asexually propagated by softwood cut-
tings to confirm the distinctness, and stability of the char-
acteristics observed in another growing season. 5 rooted
cuttings were planted in an experimental test plot located in
Viru, La Libertad, Peru. From December 2020 until the
present, said test plot has shown that the unique features of

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this new *Vaccinium* variety are stable and reproduce true to
type in successive generations of asexual propagation. In
addition, ‘BLUECSOL7’ was subsequently propagated by
softwood cutting and tissue culture and an additional plot,
with 152 plants was planted in December 2021 at the same
location. This new plot has maintained up to the present the
distinctness, and stability characteristics of the new variety
of *Vaccinium corymbosum* L.

SUMMARY OF THE VARIETY

The following is the summary of the description of the
new and distinct variety of blueberry ‘BLUECSOL7’, which
was selected in Viru, La Libertad, Peru in December 2019.
‘BLUECSOL7’ has not been observed under all environ-
mental conditions. The phenotype may vary somewhat with
variations in the environmental conditions such as tempera-
ture and light intensity, without, however, any variance in
genotype. ‘BLUECSOL7’ is an evergreen blueberry variety
with zero chill requirement. The following are the most
distinguishing traits of this new variety: strong plant vigor,
semi-upright growth habit, ovate leaf shape, globose corolla
shape and weak conspicuousness of ridges on corolla tube,
oblate fruit shaped, extra-large fruit-size, moderate bloom
intensity, medium sweetness, high acidity, and very firm
fruit.

‘BLUECSOL7’ was selected as a mid-late variety, with
high yield potential, excellent fruit quality and post-harvest
life.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying photographs show typical bush,
leaves, flower, and fruit characteristics of the new *Vaccinium*

corymbosum L. 'BLUECSOL7'. Colors shown are as accurate as can be reasonably reproduced by photographic means. Photographs were taken of 2 and 4-year-old plants grown in fields of the test plot, located in Viru, La Libertad, Peru.

FIG. 1.—Shows typical bush with upright growth habit on 2-year-old plant of the new variety 'BLUECSOL7'.

FIG. 2.—Shows the inflorescence length on 2-year-old plant of the new variety 'BLUECSOL7'.

FIG. 3.—Shows the size and shape of the flower corolla on 2-year-old plant of the new variety 'BLUECSOL7'.

FIG. 4.—Shows the upper sides and shape of complete leaves on 2-year-old plant of the new variety 'BLUECSOL7'.

FIG. 5.—Shows the underside of a complete leaf on 2-year-old plant of the new variety 'BLUECSOL7'.

FIG. 6.—Shows the size and shape of the fruits on 2-year-old plant of the new variety 'BLUECSOL7'.

DETAILED BOTANICAL DESCRIPTION

The following botanical description detailed forth distinctive traits of 'BLUECSOL7'. The data were collected from clones established in an experimental test plot in Peru on 2-year-old plants. Color descriptions are based on the Munsell® Plant Tissue Color Book scale and Pantone® color codes.

Classification:

Family.—Ericaceae.

Genus.—*Vaccinium*.

Species.—*Vaccinium corymbosum* L.

Common name.—Southern Highbush Blueberry.

Variety name.—'BLUECSOL7'.

Parentage:

Female parent.—Biloxi (public plant).

Male parent.—Unknown variety.

Plant:

Vigor.—Strong.

Growth habit.—Semi-upright.

Height.—Mean of 92 cm.

Width.—Mean of 102 cm.

Internode length.—Mean of 18.3 mm.

Evergreenness.—Evergreen.

Chilling requirements.—0 hours below 7° C.

Color of 1-year-old rough bark observed December 22.—Green Munsell® 5GY 6/6.

1-year-old wood length.—Mean of 49.5 cm.

The mature cane's color observed on July 23.—Brown Munsell® 5YR 4/4.

The mature cane's length.—About 30 to 40 cm.

Surface texture of the mature cane.—Rough.

Propagation:

Ease of propagation.—Propagates readily from soft-wood cuttings and tissue culture.

Type.—By Tissue culture.

Root description.—'BLUECSOL7' develops deep root systems and moderate volume of roots in the topsoil layer.

Rooting habit.—Moderate density and branching.

Leaves:

Length.—Mean of 56 mm.

Width.—Mean of 33 mm.

Length/width ratio.—1.73.

Shape.—Ovate.

Shape of the apex.—Acute.

Shape of the base.—Obtuse.

Surface texture in the upper.—Smooth and slightly waxy.

Surface texture in the lower.—Smooth.

Margin.—Serrate.

Color on upper side of old leaves.—Green Munsell® 7.5GY 3/4.

Color on upper side of young leaves.—Green Munsell® 5GY 5/8.

Petiole:

Length.—Mean of 4.1 mm.

Texture.—Smooth.

Color.—Green Munsell® 2.5GY 7/6.

Flowers:

Flower cluster.—Medium.

Flower fragrance.—Faint smell of geranium flowers.

Inflorescence length.—Mean of 5 cm.

Inflorescence diameter.—Mean of 2.6 cm.

Corolla:

Size of corolla tube.—Mean of 11.7 mm (from pedicel attachment point to corolla tip excluding the pedicel).

Corolla shape.—Globose.

Corolla tube color.—White Pantone® 11-0601 TPX.

Stamen:

Length.—Mean of 0.7 cm.

Number per flower.—About 9 to 10.

Color of filament.—Green Munsell® 2.5GY 8/4.

Color of anther.—Yellow Munsell® 7.5YR 6/10.

Pistil:

Number per flower.—1.

Style color.—Green Munsell® 5GY 7/10.

Style length.—Top of ovary to stigma tip. 10 mm.

Fruits:

Diameter of calyx aperture on mature berry.—Mean of 5.7 mm.

Diameter of depth of calyx on mature berry.—Mean of 2.6 mm.

Detachment force for ripe berries (easy, medium, hard).—Easy.

Berries per cluster.—About 8 to 12.

Width.—About 18 to 24 mm.

Weight.—Mean of 4.9 g per berry.

Shape.—Oblate.

Intensity of fruit bloom.—Moderate.

Pedicel scar.—Mean of 2.5 mm.

Color of unripe fruit.—Light green Munsell® 5GY 6/6.

Color of mature fruit skin with bloom.—Dapple gray Pantone® 16-3907 TCX.

Color of mature fruit skin without bloom.—Dark navy Pantone® 19-4013 TCX.

Fruit flesh color.—Green yellow Munsell® 2.5GY 8/4.

Fruit firmness.—Very firm.

Fruit acidity.—High.

Fruit sweetness.—Medium.

Market use of fruit.—Fresh market.

Seeds:

Color.—Brown Munsell® 7.5YR 5/6.

Seeds per berry.—Mean of 7 seeds.

Reproductive organs:

Pollen abundance.—High.

Pollen color.—Yellow Munsell® 5Y 7/2.

Disease, insects, and mites.—No sensitivity to any disease and pest has been observed for 'BLUECSOL7'.

Plant hardiness zone: The plants of the 'BLUECSOL7' variety were tested under the arid conditions of the northern coast of Peru, where the maximum temperatures in the year are 32° C. and the minimum are 13° C. The 'BLUECSOL7' variety can produce fruit all year round and does not require temperatures below 7° C. for flower induction. The area where its behavior was observed and evaluated does not have extreme temperatures to be able to determine cold or heat resistance.

Productivity: 'BLUECSOL7' has a super high yield potential. The harvest begins after 6 months of the planting or pruning of the plants on the field and lasts for 24 weeks. Plant productivity in the first season is about 3.5 to 4.0 kg per plant and in the second season is around 4.5 to 5.5 kg per plant. 80% of the harvest is concentrated in 17 weeks. Storage qualities: 'BLUECSOL7' has a shelf life of more than 49 days under modified atmosphere storage at 1° C. and 35 days under controlled atmosphere storage at 1° C.

COMPARISON TO THE PARENT

'BLUECSOL7' is distinguished of 'Biloxi' in the following characteristics:

Vigor plant in 'Biloxi' (Public plant) is medium and in 'BLUECSOL7' is strong.

Ratio length/width of the leaf in 'BLUECSOL7' is higher than 'Biloxi' (Public plant).

Anthocyanin coloration in the flower bud in 'BLUECSOL7' is weaker than 'Biloxi' (Public plant).

Fruit size of 'Biloxi' (Public plant) is medium, whereas in 'BLUECSOL7' fruit size large fruits.

Depth of calyx basin on mature berry is deep in 'Biloxi' (Public plant) and medium in 'BLUECSOL7'.

Diameter of calyx basin on mature berry in 'BLUECSOL7' is greater than in 'Biloxi' (Public plant)

Fruit firmness in 'BLUECSOL7' is greater than the fruit firmness in 'Biloxi' (Public plant).

COMPARISON TO COMMERCIAL VARIETY

'BLUECSOL7' is distinguished of 'Emerald' (U.S. Plant Pat. No. 12,165) in the following characteristics:

Vigor plant in 'BLUECSOL7' is greater than in 'Emerald' (U.S. Plant Pat. No. 12,165).

Leaf surface texture in the upper in 'BLUECSOL7' is smooth and slightly waxy, whereas in 'Emerald' (U.S. Plant Pat. No. 12,165) is smooth.

Shape of corolla in 'BLUECSOL7' is globose, whereas in 'Emerald' (U.S. Plant Pat. No. 12,165) is cylindric.

The abundance of pollen in 'Emerald' (U.S. Plant Pat. No. 12,165) is greater than 'BLUECSOL7'.

The margin of the leaf of 'BLUECSOL7' is serrate, whereas in 'Emerald' (U.S. Plant Pat. No. 12,165) is entire.

Fruit firmness in 'BLUECSOL7' is greater than 'Emerald' (U.S. Plant Pat. No. 12,165).

Intensity of fruit bloom in 'Emerald' (U.S. Plant Pat. No. 12,165) is weaker than 'BLUECSOL7'.

What is claimed is:

1. A new and distinct blueberry plant named 'BLUECSOL7' substantially as illustrated and described.

* * * * *



FIG 1

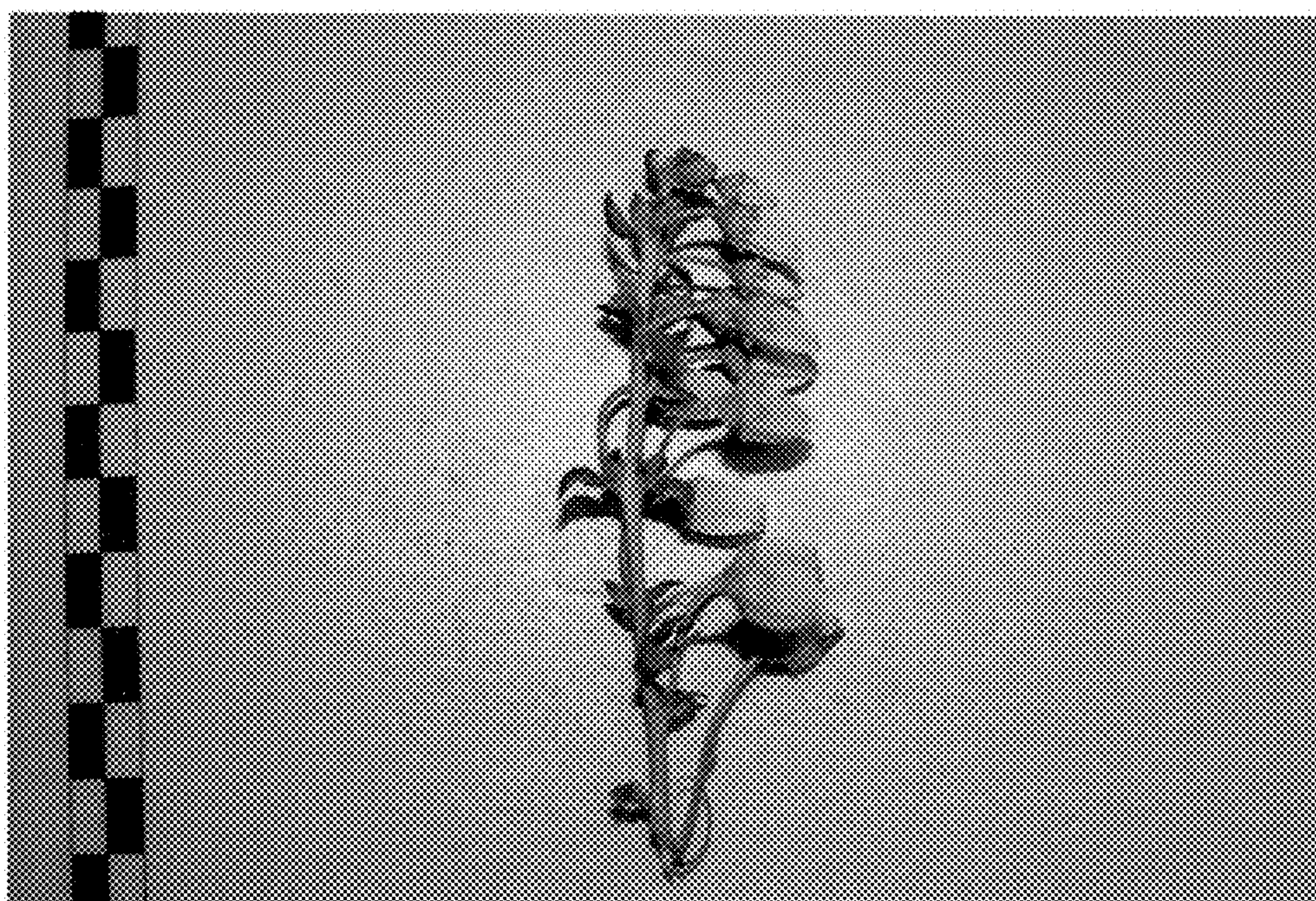


FIG 2.

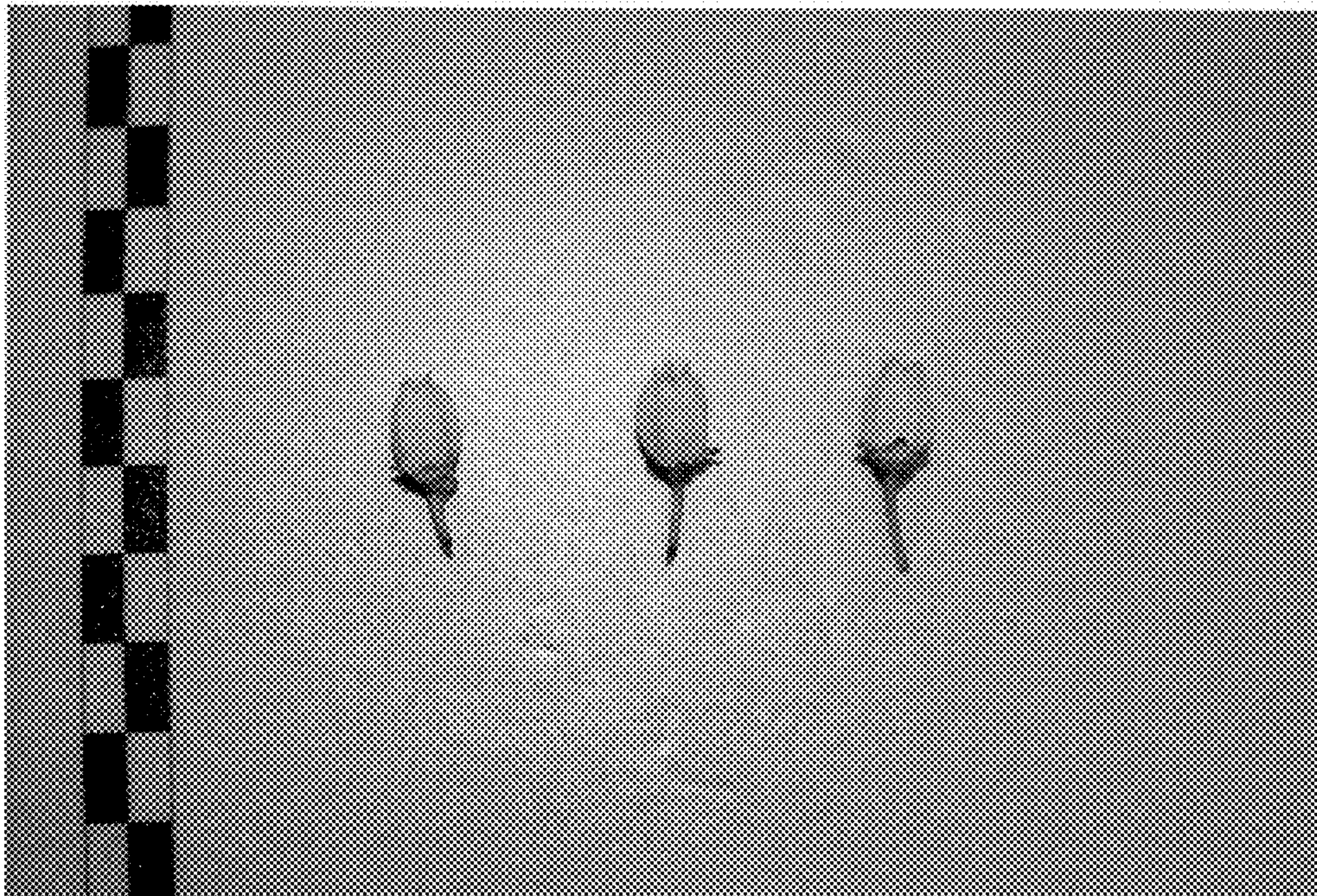


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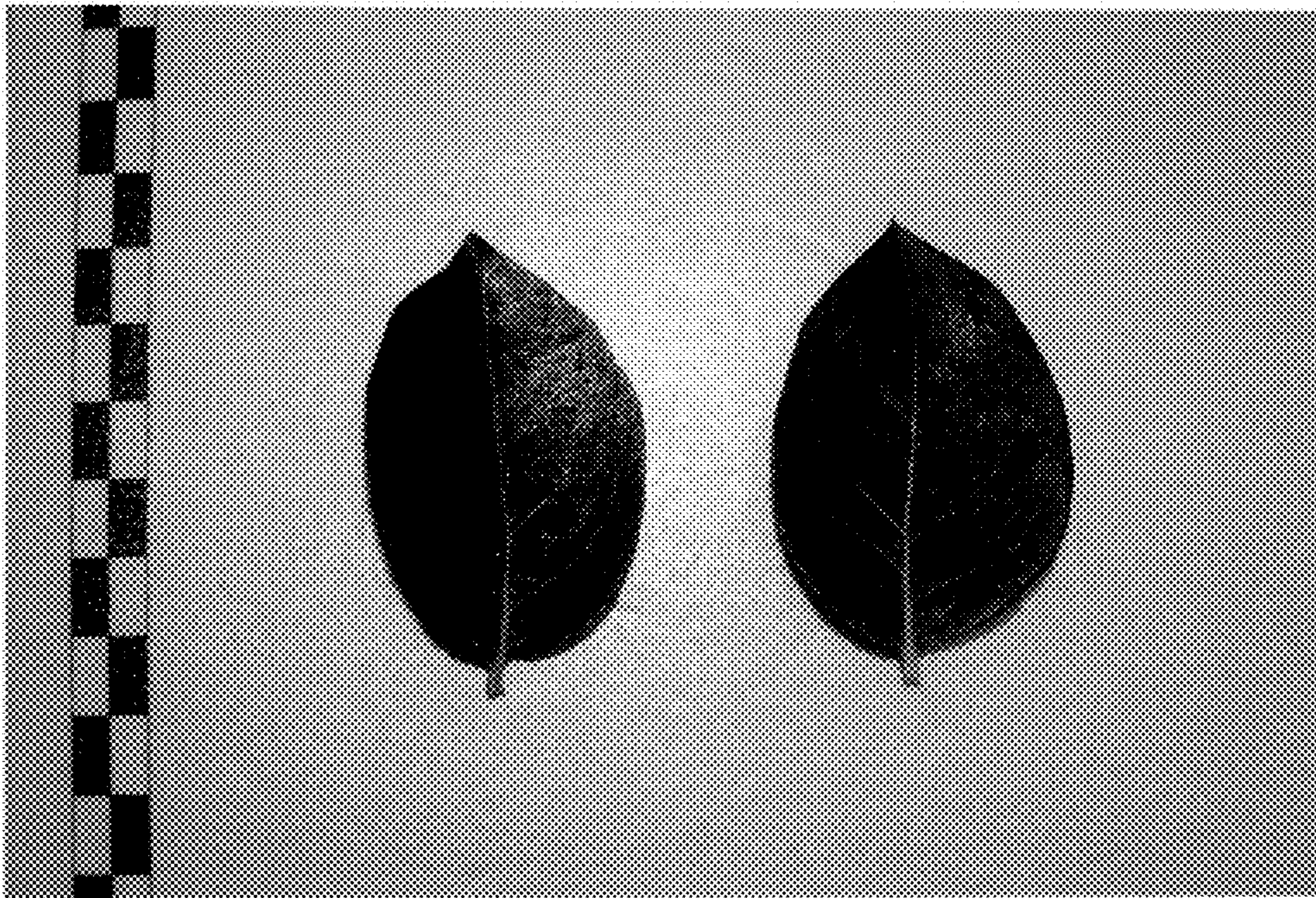


FIG. 4.

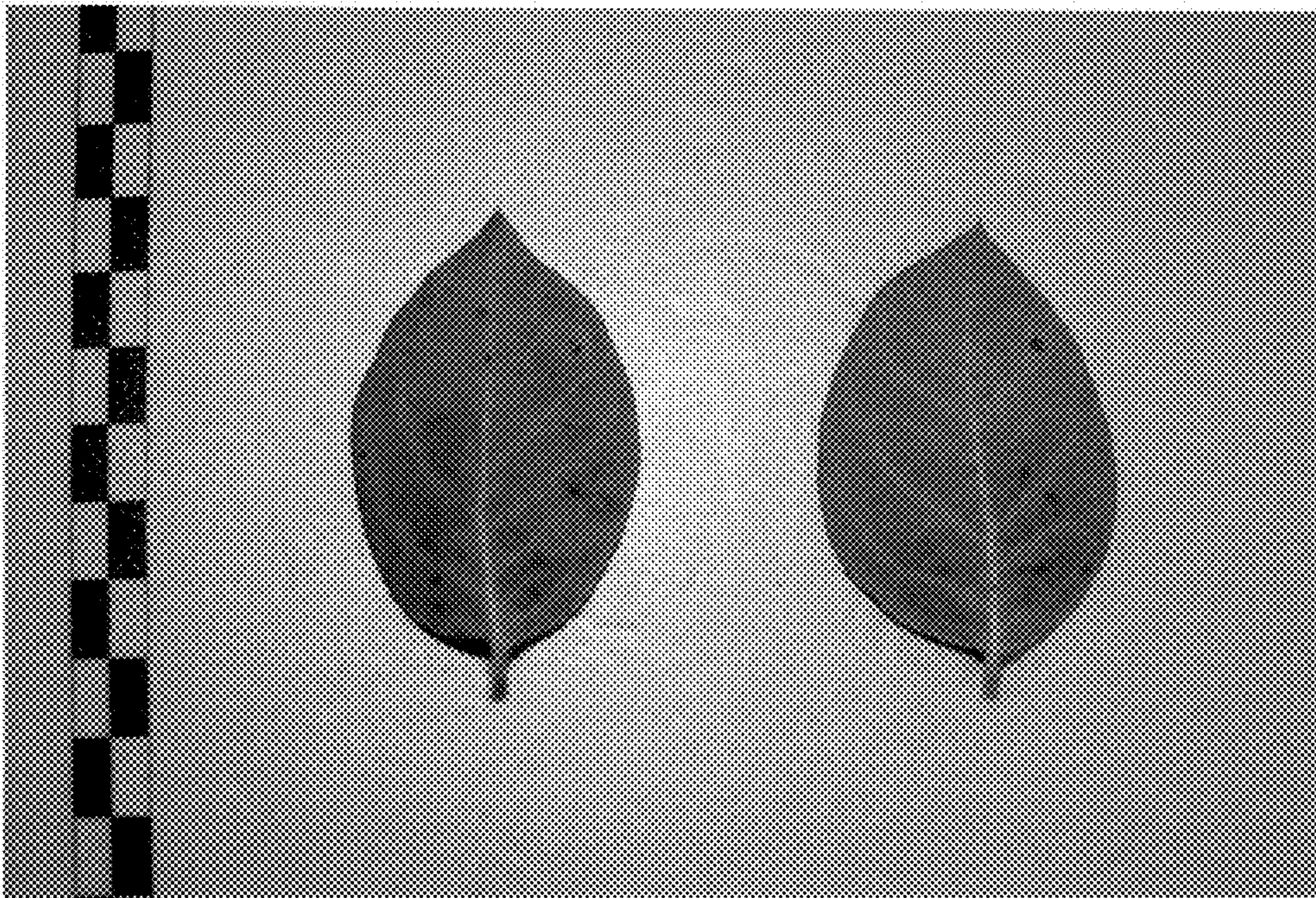


FIG 5.

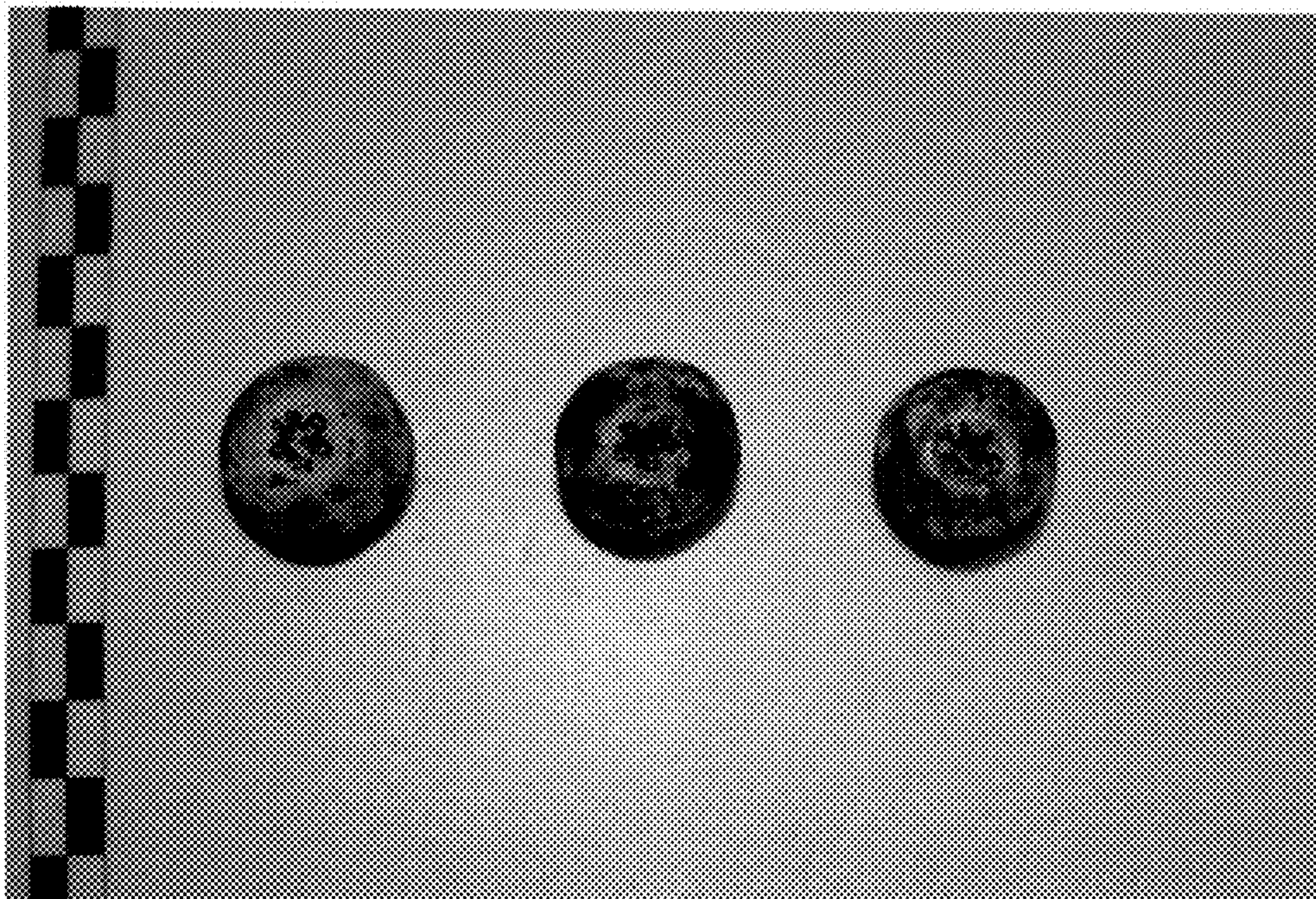


FIG. 6